CAMBRIDGE
International Education

Cambridge International A Level

MATHEMATICS **9709/31**

Paper 3 Pure Mathematics 3 **October/November 2025**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

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Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
FT	Follow through
	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
J	Judgement
JU	Judgement
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two
MR	Misread
O	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
PE	Judgment made by the PE
Pre	Premature approximation
SC	Special case
SEEN	Indicates that work/page has been seen

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Annotation	Meaning
SF	Error in number of significant figures
	Correct
TE	Transcription error
XP	Correct answer from incorrect working

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1	Commence integration by parts and reach $Ax\ln 3x + \int B\,dx$	*M1	Accept $Ax\ln 3x - B\int x \times \frac{1}{x}\,dx$. Allow unsimplified.
	Obtain $x\ln 3x - x$	A1	
	Substitute limits correctly in an expression of the form $Ax\ln 3x + Bx$	DM1	$2\ln 6 - 2 - \ln 3 + 1$
	Obtain $-1 + \ln 12$	A1	If no working seen, no marks are available.
		4	

Question	Answer	Marks	Guidance
2(a)	State or imply $2 = \log_4 4^2$	B1	
	Use log power, product or quotient law e.g. $2\log_4 (3x - 1) = \log_4 (3x - 1)^2$	M1	E.g. $\log_4 (3x - 1)^2 - \log_4 4^2 = \log_4 \frac{(3x-1)^2}{4^2}$
	Obtain a correct equation in any form, free of logs	A1	E.g. $16(2x + 1) = (3x - 1)^2$ or $9x^2 - 38x - 15 = 0$
		3	
2(b)	Solve 3-term quadratic	M1	
	Obtain answer 4.59 only	A1	$\frac{19 + 4\sqrt{31}}{9}$ OE. Accept AWRT 4.59.
		2	

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Question	Answer	Marks	Guidance
3(a)	Obtain $3\sin x + 4\cos x$	**B1**	
	State $R = 5$	**B1 FT**	FT *their* 3 and 4.
	Use trig formula to find α e.g. $\alpha = \tan^{-1}\left(\frac{3}{4}\right)$	**M1**	Using *their* 3 and 4. If using $\cos\alpha = 4$ and $\sin\alpha = 3$ seen, this scores M0A0.
	Obtain $\alpha = 36.9°$ (or awrt $36.9°$)	**A1**	
		4	
3(b)	Evaluate $\cos^{-1}\left(\frac{-4}{R}\right)$	**B1 FT**	FT *their R* from **(a)**. Allow $\cos^{-1}\left(\frac{-4}{5}\right)$ for B1 143.13…..
	Carry out a correct method to find a value of θ in the given interval	**M1**	
	Obtain answer, e.g. 60(.0)°	**A1**	Answers may be more accurate for A marks. Accept AWRT 60.0.
	Obtain a second answer, e.g. 84.6° and no other in the interval	**A1**	Ignore answers outside the given interval. Accept AWRT 84.6.
		4	

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Question	Answer	Marks	Guidance
4	Use correct product rule: $p\sin 4x\mathrm{e}^{\sin 2x} + q\cos 2x\cos 4x\mathrm{e}^{\sin 2x}$	**M1**	Condone sign error. Condone if no attempt at chain rule seen.
	Obtain derivative $-4\sin 4x\mathrm{e}^{\sin 2x} + 2\cos 2x\cos 4x\mathrm{e}^{\sin 2x}$	**A1**	OE
	Equate derivative to zero and obtain an equation in one trig function	**M1**	Allow arithmetical errors.
	Obtain $2\sin^2 2x + 4\sin 2x - 1 = 0$	**A1**	OE
	Obtain $x = 0.113$ and no other root	**A1**	May be more accurate.
		5	

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Question	Answer	Marks	Guidance
5(a)	Obtain $\text{Re}\, z \geqslant 2$	**B1**	Allow $2 \leqslant \text{Re}(z) \leqslant 3$ Condone strict inequalities throughout **(a)**. The shading is to the RHS of the diagram.
	Obtain answer of the form $\lvert z-a\rvert \leqslant b$	**M1**	Accept equation or any inequality sign. Must have a complex and b real. Condone brackets in place of modulus for this mark.
	Obtain answer $\lvert z-1+2\text{i}\rvert \leqslant 2$	**A1**	OE $\lvert z-(1-2\text{i})\rvert \leqslant 2$
		3	
5(b)	Identify the correct point	**B1**	May be clearly indicated as a point on the diagram. May be $\left(2,\ -\sqrt{3}-2\right)$. May be by attempt to solve $(x-1)^2+(y+2)^2=2^2$ with $x=2$.
	Carry out a correct method for finding the least value of arg z	**M1**	
	Obtain answer –61.8° or 298.2° or –1.08^c or 5.2(0)^c	**A1**	
		3	

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Question	Answer	Marks	Guidance
6	Use quadratic formula to solve for w	**M1**	$w = \dfrac{-4 \pm \sqrt{16 - 4(2 - \mathrm{i})(2 + \mathrm{i})}}{4 + 2\mathrm{i}}$
	Use $\mathrm{i}^2 = -1$	**M1**	
	Obtain one of the answers $w = \dfrac{-4 - 2i}{4 + 2i}$ or $w = \dfrac{-4 + 2i}{4 + 2i}$	**A1**	OE
	Multiply numerator and denominator by conjugate of *their* denominator	**M1**	Need some evidence of multiplying.
	Obtain final answers $-\dfrac{3}{5} + \dfrac{4}{5}\mathrm{i}$ and -1 [+ 0i]	**A1**	
	Alternative Method for Question 6		
	Multiply the equation by $2 - \mathrm{i}$	**M1**	
	Use $\mathrm{i}^2 = -1$	**M1**	
	Obtain $5w^2 + 4(2 - \mathrm{i})w + (2 - \mathrm{i})^2$	**A1**	OE
	Use quadratic formula or factorise to solve for w	**M1**	
	Obtain final answers $-\dfrac{3}{5} + \dfrac{4}{5}\mathrm{i}$ and -1 [+ 0i]	**A1**	

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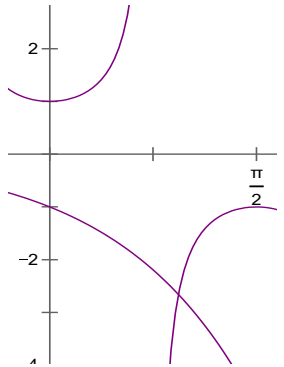
Question	Answer	Marks	Guidance
	Alternative Method 2 for Question 6		
	Substitute $w = x + iy$ and form equations for real and imaginary parts	**M1**	Two separate equations.
	Use $i^2 = -1$	**M1**	
	Obtain $2(x^2 - y^2) - 2xy + 4x + 2 = 0$ and $x^2 - y^2 + 4xy + 4y - 1 = 0$	**A1**	OE
	Form equation in x only or y only and solve	**M1**	
	Obtain final answers $-\frac{3}{5}+\frac{4}{5}i$ and -1 [+ 0i]	**A1**	
		5	

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Question	Answer	Marks	Guidance
7	State $\frac{\mathrm{d}x}{\mathrm{d}t}=2t-\frac{2}{2t+1}$	**B1**	
	Use correct quotient rule or correct product rule	**M1**	Might split first as $\frac{1}{2}-\frac{1}{4t+2}$.
	Obtain $\frac{\mathrm{d}y}{\mathrm{d}t}=\frac{(2t+1).1-t(2)}{(2t+1)^2}$ oe	**A1**	Simplifies to $\frac{1}{(2t+1)^2}$
	Use $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}t}\div\frac{\mathrm{d}x}{\mathrm{d}t}$	**M1**	Allow for $\frac{\mathrm{d}x}{\mathrm{d}y}$.
	Obtain answer $\frac{1}{2(2t+1)(2t-1)(t+1)}$	**A1**	OE. Must have 1 in numerator. May have, e.g., $\frac{1}{(2t+1)(4t^2+2t-2)}$. ISW.
	Obtain $x=\left(\frac{y}{1-2y}\right)^2-\ln\left(\frac{2y}{1-2y}+1\right)$	**B1**	OE, e.g. $x=\left(\frac{y}{1-2y}\right)^2-\ln\left(\frac{1}{1-2y}\right)$.
	Use correct quotient rule to differentiate	**M1**	
	Obtain $\frac{\mathrm{d}x}{\mathrm{d}y}=\left(\frac{2y}{1-2y}\right)\left(\frac{(1-2y)+2y}{(1-2y)^2}\right)-\frac{2}{1-2y}$	**A1**	OE, $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{(1-2y)^3}{2y-2(1-2y)^2}=\frac{(1-2y)^3}{-2(1-4y)(1-y)}$
	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\left(1-\frac{2t}{2t+1}\right)^3}{-2\left(1-\frac{4t}{2t+1}\right)\left(1-\frac{t}{2t+1}\right)}=\frac{1}{-2(2t+1)(1-2t)(t+1)}$	**M1**	Express in terms of t.
	Obtain answer $\frac{1}{2(2t+1)(2t-1)(t+1)}$	**A1**	OE
		5	

Question	Answer	Marks	Guidance
8	Separate variables correctly	**B1**	$\int \frac{kx}{x^2+1}\mathrm{d}x = \int \mathrm{e}^{-2y}\mathrm{d}y$ Condone missing integral signs or missing dx, dy, but not both.
	Obtain term $A\mathrm{e}^{-2y}$	**B1**	OE $-\frac{1}{2}\mathrm{e}^{-2y}$
	Obtain term $B\ln\left(x^2+1\right)$	**B1**	OE $\frac{k}{2}\ln\left(x^2+1\right)$
	Obtain $-\frac{1}{2}\mathrm{e}^{-2y} = \frac{k}{2}\ln\left(x^2+1\right)(+C)$	**B1**	OE
	Use $y = 0$ when $x = 0$ to evaluate c or as limits in a solution containing terms of the form e^{-2y} and $\ln\left(x^2+1\right)$	**M1**	
	Use $y = -\frac{1}{2}$ when $x = 1$ to evaluate k or as limits in a solution containing terms of the form e^{-2y} and $\ln\left(x^2+1\right)$	**M1**	
	Obtain $c = -\frac{1}{2}$ and $k = \frac{1-\mathrm{e}}{\ln 2}$	**A1**	OE, e.g. $-\frac{1}{2}\mathrm{e}^{-2y} = \frac{1-\mathrm{e}}{2\ln 2}\ln\left(x^2+1\right) - \frac{1}{2}$.
	Obtain final answer $y = -\frac{1}{2}\ln\left(2\mathrm{e}-1\right)$	**A1**	Or exact equivalent.
		8	

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Question	Answer	Marks	Guidance
9(a)	Sketch a relevant graph, e.g. $y = \sec 2x$ May be sketching $\cos 2x$ and $-e^{-x}$  For $y = \sec 2x$ correct shape, asymptote in correct position. Cuts at (0, 1) For $-e^{x}$ correct shape, cuts at (0, 1) For $y = \cos 2x$ correct shape, correct max/min, cuts axis at $\frac{\pi}{4}$ For $-e^{-x}$ correct shape, cuts at (0, -1)	**B1**	 Complete scale not needed, but should have enough to imply the key points for both graphs.
	Sketch a second relevant graph, e.g. $y = -e^{x}$ and justify the given statement	**B1**	Needs to mark intersection with a dot, a cross, or say roots at points of intersection, OE.
		2	
9(b)	Calculate the values of a relevant expression or pair of expressions at $x = 0.9$ and $x = 1$	**M1**	E.g. $\begin{matrix}-4.401 < -2.460 \\ -2.403 > -2.718\end{matrix}$ or $\begin{matrix}-1.94 < 0 \\ 0.315 > 0\end{matrix}$ (or $\begin{matrix}0.179 > 0 \\ -0.048 < 0\end{matrix}$ for the reciprocal graphs).
	Complete the argument correctly with correct calculated values	**A1**	
		2	

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Question	Answer	Marks	Guidance
9(c)	State the equation $x = \frac{1}{2}\cos^{-1}\left(-e^{-x}\right)$ and rearrange to obtain sec $2x = -e^{x}$ Or rearrange sec $2x = -e^{x}$ as $x = \frac{1}{2}\cos^{-1}\left(-e^{-x}\right)$ and state $x_{n+1} = \frac{1}{2}\cos^{-1}\left(-e^{-x_n}\right)$	**B1**	
		1	
9(d)	Use the iterative process correctly at least once	**M1**	
	Obtain final answer 0.978	**A1**	
	Show sufficient iterations to at least 5 d.p. to justify 0.978 to 3 d.p. or show there is a sign change in the interval (0.9775, 0.9785)	**A1**	E.g. 1, 0.97376, 0.97903,0.97796, 0.97813 0.95, 0.98395, 0.97697, 0.97838, 0.97809 0.9, 0.99475, 0.97480, 0.97882, 0.97800, 0.97817
		3	

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Question	Answer	Marks	Guidance
10(a)	State or imply the form $A+\frac{B}{3+x}+\frac{Cx+D}{2+x^2}$	**B1**	
	Use a correct method for finding a constant	**M1**	Might be working from $1+\frac{-3x^2-17}{(3+x)(2+x^2)}$.
	Obtain one of $A = 1$, $B = –4$, $C = 1$ and $D = –3$	**A1**	**SC**: If B0 scored due to missing term(s), then a maximum of M1A1 is available for a correct method leading to a correct value.
	Obtain a second value	**A1**	**SC** if obtaining $A = 1$ and then scoring B0, they can score maximum M1A1 + A1 for a correct value for one other constant.
	Obtain a third value	**A1**	
	Obtain all four correct values	**A1**	
		6	

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Question	Answer	Marks	Guidance
10(b)	Use a correct method to find the first two terms in the expansion of $(3 + x)^{-1}$, $\left(1+\frac{x}{3}\right)^{-1}$, $(2 + x^2)^{-1}$ or $\left(1+\frac{x^2}{2}\right)^{-1}$	**M1**	Symbolic binomial coefficients not sufficient for the M1.
	Obtain correct unsimplified expansions up to the term in x^2 of each partial fraction	**A1 FT**	FT *B*, *C* and *D*.
		A1 FT	E.g. $\frac{B}{3}\left(1-\frac{x}{3}+\frac{x^2}{9}\right)$ and $\frac{Cx+D}{2}\left(1-\frac{x^2}{2}\right)$.
	Multiply $(Cx + D)$ by the expansion of $\left(1+\frac{x^2}{2}\right)^{-1}$ up to the term in x^2 where $CD \neq 0$	**M1**	Expansion of $\left(1+\frac{x^2}{2}\right)^{-1}$. Must be of the form $1 + Qx^2$. Multiplication must include 3 relevant terms.
	Obtain final answer $-\frac{11}{6}+\frac{17}{18}x+\frac{65}{108}x^2$	**A1**	Or exact equivalent. Do not ISW attempts to multiply expression by, e.g., 108.
		5	

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Question	Answer	Marks	Guidance
11(a)	Carry out a correct method for finding a vector equation for m	**M1**	
	Obtain $\mathbf{r} = \begin{pmatrix}1\\5\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\2\end{pmatrix}$	**A1**	Or $\mathbf{r} = \begin{pmatrix}0\\4\\1\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\2\end{pmatrix}$ OE. Must have $\mathbf{r} = \ldots$
		2	
11(b)	Express general point of the second line in component form, i.e. $(1 + 2\mu, -3 - 2\mu, 1 + 3\mu)$	**B1**	Or $(3 + 2\mu, -5 - 2\mu, 4 + 3\mu)$.
	Equate at least two pairs of corresponding components and solve for λ or for μ	**M1**	
	Obtain $\lambda = -4$ or $\mu = -2$	**A1**	Note if m has direction vector $\begin{pmatrix}-1\\-1\\-2\end{pmatrix}$, then $\lambda = 4$. If line through CD has direction vector $\begin{pmatrix}-2\\2\\-3\end{pmatrix}$, then $\mu = 2$.
	Obtain position vector of point of intersection is $\begin{pmatrix}-3\\1\\-5\end{pmatrix}$	**A1**	Accept coordinates.
		4	

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Question	Answer	Marks	Guidance
11(c)	Find $\overrightarrow{CP}$ for a general point P on m, e.g. $\begin{pmatrix}0\\8\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}$	**B1**	Or $\begin{pmatrix}-1\\7\\0\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}$.
	Calculate scalar product of $\overrightarrow{CP}$ and a direction vector for m and equate the result to zero	**M1**	
	Obtain $\lambda = -2$	**A1**	OE
	Obtain answer $\begin{pmatrix}-1\\3\\-1\end{pmatrix}$	**A1**	OE
		4	

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